CarePath AI — Delivery Plan by Phase

Source of truth	CarePath-AI-Solution-Plan.pdf (22 Oct 2026). This plan turns it into executable steps; where they differ, the Solution Plan wins until the change is agreed at a sync and logged in the decision register (§11).
UI reference	CarePath-AI-UI-demo-srpski.html, screens for demo steps 1–10
Horizon	4 workshop weeks: preparation, spec + simulator, one build week, stabilisation
Constraint	Zero cost and fully offline on the day of the demo
Last updated	27 Oct 2026

1. Summary

We deliver a working PoC on 10,000 synthetic heart-failure patients, optionally plus about 3,000 with type 2 diabetes, and a live demo of about 7 minutes for an Italian audience. Everything runs locally on free, open-source software and free local models.

The plan has five phases. Each one ends with a gate, measured against the acceptance criteria (AC1–AC19) from the Solution Plan:

Phase	When	Goal	Exit gate
0. Foundations	Week 1	Same environment for everyone, models chosen on numbers, repo skeleton running	Hello stack runs from a clean clone; model decision recorded
1. Spec, contracts, simulator	Week 2	Frozen contracts; simulator with 3 scenarios and Maria that provably contains the injected patterns	**Scope gate:** pattern notebook passes for all 3 scenarios, Maria trajectory stable (AC13 data side, AC14), SPEC approved
2. Core build	Week 3, days 1–5	Whole demo flow works end-to-end every day	Core ACs measurable: AC1–AC8, AC13, AC14, AC17
3. Gated extensions	Week 3, days 3–5 (parallel)	Optional agents, option B, Bed Finder, added in order and only if the core is healthy	Checkpoint on day 3 and AC10–AC12, AC15, AC16, AC18, AC19
4. Stabilisation and demo	Week 4	10 consecutive offline runs without a stop	AC9; runbook rehearsed on the main and the backup machine

The biggest risk is unchanged from the Solution Plan: the simulator in week 2. The biggest new finding is that local models on our laptops are slow and occasionally wrong about facts (§3). So the design is **cache-first**: every LLM output that appears in the demo is generated before the demo and replayed, and the demo starts in cache-only mode (D14). Live generation is a bonus, not a dependency.

2. Engineering principles

These apply to every phase. A pull request that breaks one of them needs a decision-register entry.

1. **Contracts before code.** Every interface between roles is in `spec/SPEC.md` before anyone builds against it. Whoever waits on another role works with fixtures of the same shape (`testing/fixtures/`).
2. **Code computes, the LLM phrases.** Numbers, names, dates and rankings come from code (rules, SQL views, XGBoost). The LLM only writes sentences around values it received from a tool or a template. It never produces a fact that a tool did not return.
3. **Cache-first LLM.** Seeds are fixed, so the inputs at every rehearsal are identical. Every LLM call goes through one gateway (§7.2, step 2.7), keyed by `(agent, model, prompt_version, input_hash)`. The cache is warmed ahead of

time on the demo machine (D12). Only outputs the demo shows are cached; every other text comes from its template (D17).

4. **One writer.** Agents only read, through tools. The action service is the only component that changes state, and only after the autonomy and guardrail checks. Every action is written to the audit log and can be undone.
5. **Disease is configuration.** Pathway JSON plus data, never `if pathway == "hf"` in code (AC17).
6. **Offline by default.** No CDN, no public APIs, no telemetry. Fonts, JS, GeoJSON, models and cache all live on disk. We test with the network switched off from week 3.
7. **Zero cost.** Only tools with permissive or free licences and no paid tier in the critical path (§10).
8. **End-to-end every day from week 3.** A feature that does not work in the full flow is not done.

3. Environment baseline (measured 26 Oct 2026)

Item	State
Reference laptop (every team laptop, D12)	Intel i7-1355U (10 cores, 12 threads), 32 GB RAM, Intel Iris Xe, **no NVIDIA GPU**
Python	3.12 via `uv`; `.venv` from `uv sync` (`pyproject.toml`, `uv.lock`)
Packages	numpy, pandas, duckdb, scikit-learn, xgboost, fastapi, uvicorn, pydantic, ollama; dev: pytest, jupyterlab
LLM runtime	Ollama 0.40.11, local server on `127.0.0.1:11434`
Models pulled	`qwen3:30b-instruct` (18 GB), `qwen3:4b-instruct` (2.5 GB), D11. Evaluated and deleted from disk (results kept in `testing/results/`): `nemotron-3.5-lightning:30b`, `qwen3:8b`, `qwen2.5:7b`, `qwen2.5:14b`, `mistral-nemo:12b`
Evaluation	`testing/llm_eval.py <model> [...]`; `--investigate-only` reruns just the investigation; results in `testing/results/`

3.1 Model evaluation (step 0.4)

Seven free local models, CPU only, temperature 0, seed 42, thinking off. Three tests per model:

- **Italian:** 5 texts written from facts that code supplies (MMG note, alarm, call script, rule proposal, report line), checked automatically for missing and wrong facts, then read by a person.
- **Tools:** 5 questions with the 7 tools of Solution Plan §7; correct tool, valid schema, expected arguments.
- **Investigation:** the Investigator flow with fake tool results: several tool calls, then a JSON answer that must contain the key facts (41 days at Ospedale B, 34 free slots at Ospedale C, Cooperativa 3 at 52%).

Model	Licence	RAM	Speed	Italian	Tools	Investigation (v2 prompt)	Verdict
`qwen3:30b-instruct` (MoE, 3B active)	Apache 2.0	18 GB	10.8 tok/s	**5/5**, 11.5 s each	4/5, 18 s	**Pass**, 243 s, all facts right	**Recommended: all cached agents**
`qwen3:4b-instruct`	Apache 2.0	2.5 GB	**14.3 tok/s**	4/5, 8.1 s each	3/5, 15 s	Pass, 244 s	**Recommended: short live text, behind templates**
`nemotron-3.5-lightning:30b` (MoE, 3B active)	NVIDIA Open Model	25 GB	5.6 tok/s	5/5, 24.4 s	4/5, 67 s	Pass, 394 s	Backup only: too heavy for 32 GB next to the app
`qwen3:8b`	Apache 2.0	5.2 GB	5.6 tok/s	4/5	5/5	Fail (v1): invented an ASST, missed key facts	Not recommended: slower and weaker than 30B MoE

Model	Licence	RAM	Speed	Italian	Tools	Investigation (v2 prompt)	Verdict
`qwen2.5:7b`	Apache 2.0	4.7 GB	5.7 tok/s	4/5; wrote "Medico di Maggiore Grado" for MMG	5/5	Fail (v1): invented an ASST	Not recommended
`qwen2.5:14b`	Apache 2.0	9.0 GB	3.2 tok/s	4/5; **inverted Maria's fact** ("continua la terapia con furosemide")	5/5	Fail (v1), 408 s	Not recommended
`mistral-nemo:12b`	Apache 2.0	7.1 GB	3.6 tok/s	5/5	4/5	Fail (v1): **called no tools**, answered from nothing	Not recommended

How to read it:

- **No model is fast enough live for the Investigator on our laptops.** The best investigation took 243 s against the 30 s target (AC5). Every investigation shown in the demo comes from the cache (D12, D14).
- **The 30B MoE model beats every dense model** because only 3B parameters are active per token. It is faster than the 7B and 8B dense models and is the only one with every Italian text right.
- **The prompt matters as much as the model.** With the first summary prompt ("riassumi in JSON"), 6 of 7 models dropped the 34 free slots at Ospedale C, which is the fact the reschedule action needs. With a prompt that asks for every relevant number, all three candidates passed. Prompts are versioned in the gateway (`prompt_version`) for this reason.
- **Every model made at least one factual slip somewhere:** inverted meaning, a wrong hospital in a call script, an invented ASST, "within 30 days" instead of 21, or adherence presented as the cause of the wait. Facts come from code (principle 2), and the Verifier stays in scope.
- **One question is ambiguous, not wrong:** every model read "per gruppo di MMG" as `grouping=mmg` instead of `mmg_group`. Fix the question or the tool description before scoring AC6.
- **Speeds vary by about ±40% on the same laptop** with load and power mode (7B measured 9.0 and 5.7 tok/s on two runs). Measure on mains power with nothing else running.

Limits of this evaluation: one run per prompt and 5 prompts per test, on fake tool results. AC6 (10 benchmark questions × 3 scenarios on real simulator data) is the real gate in week 3.

4. Roles

Role	Owns	Accepts
R1 Data & simulator	simulator, 3 scenarios, Maria, calibration, diabetes data (option B)	AC8, AC13, AC14
R2 Analytics & ML	state engine, views, Risk, what-if, PNE forecast, Enrollment rules, Bed Finder ranking	AC2–AC4, AC12, AC15–AC18
R3 AI agents	all LLM agents, LLM gateway, cache, action service, autonomy, Verifier, prompt log	AC1, AC5–AC7, AC10, AC11, AC19
R4 Console	all screens, WebSocket client, demo bar, offline assets	supports AC9
R5 Spec, domain & demo	SPEC.md, contracts, PDTA texts, Italian terminology, demo script, runbook, presentation	AC9, AC ordering

The critical path is R1 and R5 in week 2.

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5. Phase 0 — Foundations (Week 1)

Goal: everyone can run the same stack from a clean clone, the models are chosen on measured numbers, and the shape of the system is agreed.

Entry: repo exists (done), Solution Plan approved.

#	Step	Owner	Output	Done when
0.1	Team charter: roles R1–R5, two 20-min syncs a week, chat channel, decision-register owner (R5)	All	`brief/charter.md`	Signed off by all five
0.2	Approval check: confirm with the mentor that Ollama, the chosen models and every tool in §10 are allowed. No IT approval needed: synthetic data only, and Ollama and the models already run on the team laptops (§3)	R5	Line in decision register	Written "yes" from the mentor
0.3	Dev setup guide: `uv sync`, Ollama install, `ollama pull`, PATH fix for terminals opened before install, run the benchmark	R3	`README.md` "Getting started"	A second person follows it on a clean laptop without help
0.4	Model selection: run `llm_eval.py` (5 Italian, 5 tool-calling prompts, one investigation) on the candidates with no other load; R5 reads the Italian outputs for factual faithfulness. **Done on the reference laptop (§3.1); D11 decided 27 Oct 2026.** Follow-up checks, not blockers: R5 review of the Italian outputs, and one rerun on the backup laptop	R3, R5	Table of tok/s, tool-call accuracy and faithfulness per model; decision D11	One small model (live text, Daily Worklist) and one larger model (cache generation for Investigator, Pathway Builder, Rule Advisor) chosen
0.5	Demo machine decision. **Decided (D12):** every team laptop is the reference laptop, none has a GPU. Remaining: name the demo machine (it also generates the cache) and the backup laptop	R5, R3	Decision D12	Two named machines
0.6	Repo skeleton exactly as in Solution Plan §15: `brief/ spec/ src/{sim,engine,agents,api,ui}/ testing/ demo/`; `sim`, `engine`, `agents` and `api` are separate Python packages installed by `uv sync`; `carepath-api` entry point. **Done**	R4	Skeleton PR	`uv run pytest` passes
0.7	API skeleton: FastAPI app, `GET /api/health`, WebSocket `/api/stream` that emits a fake tick every 2 s; serves the built console. **Done**	R4	`src/api/main.py`	Browser shows a ticking counter
0.8	Frontend decision. Streamlit is not recommended because it cannot reproduce the prototype layout. **Decided (D13):** React 19 + TypeScript on Vite 8, Node 24 via fnm; console shell in `src/ui`	R4	Decision D13	Recorded with reason
0.9	DuckDB skeleton: database file layout, empty tables from the draft contracts, state-engine stub	R2	`src/engine/db.py`	Tables created by one command
0.10	Simulator sketch: list of comuni in ASST Mantova, Cremona and Crema; ISTAT population by age and sex; published PNE 30-day HF readmission range; Ospedale A/B/C mapped to ASSTs	R1	`src/sim/README.md` with sources and numbers	Reviewed by R5
0.11	Problem brief and SPEC draft: users, pains, success metrics; draft of every contract in §6 of this plan. SPEC draft v0.1 done; `brief/problem.md` open	R5	`brief/problem.md`, `spec/SPEC.md` draft	Reviewed at the second sync

Exit gate: a clean clone runs the hello stack and the benchmark. Models and the demo machine are chosen with numbers (D11, D12). SPEC draft has been reviewed.

Risks: No GPU is already a fact (D12) and is handled: CPU only, cache-only demo (D14), first cache warm-up in week 3, day 3 (step 2.11).

6. Phase 1 — Spec, contracts and simulator (Week 2, scope gate)

Goal: contracts are frozen, and the simulator for heart failure, with 3 scenarios and Maria, provably contains patterns 1–3, is calibrated to PNE, and runs a day in under 1 s.

Entry: Phase 0 exit gate passed.

6.1 Spec and contracts (R5 leads, all review)

#	Step	Owner	Output	Done when
1.1	Data contracts: `events` (11 columns per Solution Plan §6), `patients`, `facilities`, `capacity_slots`, `beds` (optional), `scenario_truth`; types, keys, allowed codes	R5, R1	SPEC §Data	R1 and R2 sign off
1.2	State and view contracts: `pathway_state` (one row per patient × pathway; status `on_track` / `due_soon` / `overdue` / `dropped`), `kpi_daily`, `sankey_flows`, `map_comuni`, `worklist_today`	R5, R2	SPEC §State	R2 and R4 sign off
1.3	Pathway model JSON schema plus `hf_lombardia_v1.json` (eligibility, steps, due/every days, rules); codes confirmed by the Italian colleague	R5, R2	`spec/pathways/`	Validates against the schema in a test
1.4	Tool signatures (Solution Plan §7): all take `pathway_id`, read-only, typed return JSON; the error contract	R5, R3	SPEC §Tools	R3 signs off
1.5	Action contract: types (`notify_mmg`, `reschedule`, `call_task`, `enroll_proposal`, `rule_proposal`, `pathway_proposal`, `bed_proposal`, `bed_assignment`), autonomy levels 0–3, status lifecycle, evidence list	R5, R3	SPEC §Actions	R3 signs off
1.6	API and WebSocket contract: REST endpoints from Solution Plan §7; WS envelope `{type: tick|alert|action|investigation, scenario, day, payload}`	R5, R4	SPEC §API	R4 signs off
1.7	Acceptance criteria AC1–AC19 copied into SPEC with owner, measurement method and test file name	R5	SPEC §Acceptance	Each AC has a test stub in `testing/`
1.8	PDTA texts in Italian: heart failure (final), type 2 diabetes (draft)	R5	`spec/pdta/`	Reviewed by the Italian colleague
1.9	**Contract freeze** at the end of day 2; later changes only via the sync and the decision register	R5	Tag `spec-v1`	Tag pushed

6.2 Reference data (R1)

#	Step	Output	Done when
1.10	Download ISTAT comune list, population by age band and ISTAT code for the 3 ASSTs; store as CSV with source URL and licence	`src/sim/data/comuni.csv`	Totals match ISTAT for each ASST
1.11	Download comune boundaries, clip to the 3 ASSTs, simplify to under 2 MB GeoJSON for the offline map	`src/ui/assets/comuni.geojson`	Renders offline in a test page
1.12	Facility table: Ospedale A, B, C and clinics, placed in real ASSTs, with capacity per service (echo, cardiology visit, lab)	`facilities.csv`	Reviewed by R5
1.13	Distance matrix comune → facility, precomputed once (haversine × road factor ~1.3; OSRM only if accuracy is challenged); never queried live	`distances.parquet`	Maria's comune shows about 45 km to her hospital

6.3 Simulator (R1)

#	Step	Output	Done when
1.14	Population generator: 10,000 HF patients sampled by comune and age; lives alone; distance; MMG assignment; MMG grouped into cooperatives with a Centro Servizi	`patients` table	Distributions plotted in the notebook
1.15	Daily vectorised step (numpy and pandas, fixed seed): discharges, visits, imaging, labs, dispensing, telemonitoring readings, admissions; capacity-driven waits from `capacity_slots`	`src/sim/step.py`	365 days for 10k patients run; one day under 1 s (early AC2)
1.16	Outcome model: readmission hazard rising with non-collection, missed visits, long waits, distance and living alone; calibrated so 30-day HF readmission falls inside the published PNE range	Config with every assumed probability	Notebook shows the rate inside the range
1.17	Patterns 1–3 parameterised per scenario: echo bottleneck at one hospital with free slots at another; one MMG group with low furosemide collection; rural cluster (lives alone, more than 40 km) dropping telemonitoring	`src/sim/scenarios/{1,2,3}.yaml`	Same pattern types, different locations per scenario
1.18	`scenario_truth` written to a separate path; agent tools are given a DuckDB connection without that table; a test asserts it	`demo/truth/` + test	Test fails if any tool can read it
1.19	Maria: scripted trajectory present in all 3 scenarios; without agents she is readmitted on day 47; with agents a day-31 call leads to delivery on day 33, HbA1c on day 40 and no readmission	Maria overrides + test	AC14 passes 10 runs out of 10
1.20	Action feedback hook: an executed action on day D becomes events on day D+1 (interface plus a no-op implementation)	`src/sim/feedback.py`	R3 can call it
1.21	Pattern-proof notebook: for each scenario shows patterns 1–3 and the PNE calibration from data alone	`testing/notebooks/patterns.ipynb`	R5 reviews; this is the gate evidence

6.4 Parallel work in week 2

#	Step	Owner	Output
1.22	State engine over simulator output: incremental `pathway_state` update for patients with new events plus a deadline sweep; `kpi_daily`	R2	`src/engine/state.py`
1.23	Fixtures for every contract (one small scenario) so R3 and R4 never wait	R2	`testing/fixtures/`
1.24	Tools v1 against fixtures; 10 benchmark questions with expected answers per scenario	R3	`src/agents/tools.py`, `testing/benchmark/`
1.25	Screens on fixtures: Storia di Maria, Buste, Panoramica, Attività agenti side panel	R4	`src/ui/`

Scope gate (end of week 2): the pattern notebook passes for all 3 scenarios, Maria is stable (AC14), one simulated day runs in under 1 s, and SPEC `spec-v1` is tagged. **If it fails:** option B (diabetes) moves to a slide immediately; generic contracts stay; R1 keeps working on the simulator in week 3 and R2 takes the state engine alone.

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7. Phase 2 — Core build (Week 3, days 1–5)

Goal: the complete demo flow (steps 1–7, 9, 10) runs end-to-end every day on real simulator data, with the cache in place.

Entry: scope gate passed, or failed with the cut applied.

Daily rule: at the end of each day, R5 runs the whole flow on the main branch. Whatever breaks is the first task the next morning.

7.1 Engine and analytics (R2)

Day	Step	Output	Done when
1	2.1 Real-time tick loop in the order of Solution Plan §9: sim emits day D → state update → Risk on changed patients → Monitoring → refresh views → Intervention → Daily Worklist → Investigator (async) → WS push	`src/engine/tick.py`	1 day under 1 s for 10k patients (AC2)
1	2.2 Views `sankey_flows`, `map_comuni`	DuckDB views	Under 200 ms each (AC3)
2	2.3 Risk model: XGBoost trained once offline on a separate seed; model file versioned; top-3 drivers per patient	`src/engine/risk.py`, `models/risk.json`	AUC reported in notebook; scoring on a tick within budget
2	2.4 `worklist_today`: one row per patient across pathways, ranked by risk, all reasons, escalation step	View + test	Top 20 contain at least 15 high-risk or overdue patients (AC12 data side)
3	2.5 What-if: run the simulator forward 6 months with changed levers (echo capacity, telemonitoring deadline), with and without agents, same seed	`src/engine/whatif.py`	Under 10 s (AC4)
5	2.6 PNE forecast to year end, with and without CarePath, plus a lever	`src/engine/pne.py`	Numbers shown on step 7 come from here

7.2 Agents and control (R3)

Day	Step	Output	Done when
1	2.7 **LLM gateway**: one function used by every agent; temperature 0, seed, JSON schema via Ollama structured output, 1 retry, 20 s timeout in the demo and 600 s in warm-up runs (one 30B investigation takes about 4 min, §3.1), `keep_alive` so the model stays loaded, cache key `(agent, model, prompt_version, input_hash)`, cache-only switch, one JSON log line per call (agent, model, latency, cache hit, valid)	`src/agents/llm.py`, `demo/cache/`, `logs/llm.jsonl`	Unit tests for hit, miss, timeout and invalid JSON
1	2.8 Monitoring (rules from pathway JSON; regional alarm when the same deviation exceeds a threshold at one facility) and Risk wired to the tick	`src/agents/monitoring.py`	Echo alarm for Ospedale B fires around day 35 in scenario 2
2	2.9 Investigator: tool-only loop, max 6 steps, emits a JSON list of claims, each with the tool call that supports it; no SQL	`src/agents/investigator.py`	Finds pattern 1 in all 3 scenarios without hints (AC1 part)
3	2.10 Benchmark run: 10 questions × 3 scenarios; scoring script; results stored per prompt version	`testing/benchmark/run.py`	At least 8 of 10 correct (AC6)
3	2.11 Cache warm-up v1 on the demo machine for all 3 scenarios with `qwen3:30b-instruct` (D11), unattended, on mains power with nothing else running. First list the outputs the demo script shows (D17), count them and estimate the run time at the §3.1 speeds	Cache manifest with counts and estimated time; cache files	Investigator from cache under 2 s (AC5); every manifest entry is in the cache
4	2.12 **Action service**: single writer; autonomy level per action type; guardrails from Solution Plan §8 (no therapy, no diagnosis, daily limit per facility); audit log; undo; feeds the simulator via the feedback hook	`src/agents/actions.py`	No action above its level executes without approval (AC7)

Day	Step	Output	Done when
4	2.13 Intervention: rules choose the action (reminder, call task, notify MMG, reschedule to the earliest free slot then distance); every rationale is a template filled by code; LLM wording only for the actions the demo shows (D17)	`src/agents/intervention.py`	With-agents run has fewer readmissions on the same seed (AC8)
5	2.14 Reporter: monthly Italian report in PNE style; numbers from views, LLM for sentences only	`src/agents/reporter.py`	Report numbers equal the view numbers (test)
5	2.15 Pathway Builder lite: Italian PDTA text → pathway JSON → schema validation → retry → manual JSON fallback	`src/agents/pathway_builder.py`	HF PDTA reproduces `hf_lombardia_v1.json`

7.3 Console (R4)

Day	Step	Done when
1	2.16 WebSocket client; Mappa from local GeoJSON (fallback: comune table, as in the prototype); Percorsi (Sankey)	Map recolours each tick, offline
2	2.17 Chiedi (investigation steps appear one by one, "verificato" badge slot); Paziente 360 (timeline, with/without toggle)	Driven by real API data
3	2.18 Impostazioni (autonomy per action type); Approvazioni (queue, approve and reject call the API)	Approve in UI changes the audit log
4	2.19 What-if screen (sliders, run, results); Confronto split screen with counters	Matches §7.1 outputs
5	2.20 Lista chiamate; **demo bar** (hidden; hotkeys for scenario 1/2/3, pause, speed, jump to day, cache-only, reset); cache-only is on at start (D14); cache indicator visible to the team only	Operator can run the whole script with hotkeys

7.4 Spec, tests and demo (R5)

Day	Step
1	2.21 Acceptance tests from AC stubs, run locally with `uv run pytest -m acceptance`
2	2.22 Check AC1–AC3 on real data; file defects
3	2.23 **Checkpoint:** core runs end-to-end? Decide which optional layers enter (Phase 3)
4	2.24 Demo script v1: trigger sentences, operator clicks, timings from Solution Plan §11
5	2.25 Technical review with the mentor

Exit gate (end of week 3): steps 1–7, 9 and 10 of the demo run end-to-end on simulator data with the cache. AC1–AC8, AC13, AC14 and AC17 have passing or measured tests.

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8. Phase 3 — Gated extensions (Week 3, days 3–5, in parallel)

Entry: checkpoint on day 3 says the core runs end-to-end. Extensions are added **in this order**. Each one is cut as a whole if it is not demo-ready by the end of week 3.

Order	Extension	Steps	Owner	AC	Demo step
1	Verifier	Re-run the tool behind each Investigator claim; compare with tolerance; label `verificato` or correct; test set of 5 answers with injected wrong numbers	R3	AC10	5
2	Rule Advisor	From an investigation finding, propose a rule in pathway-rule format; on approval Monitoring loads it the next day	R3	AC11	5

Order	Extension	Steps	Owner	AC	Demo step
3	Daily Worklist	Ranking from `worklist_today` (code); Italian call script for the top 20 only, from a template, with LLM wording cached for the days the demo shows (D17); call outcome field	R2, R3	AC12	9
4	Option B: diabetes	R1: diabetes population (~3,000, ~900 overlapping), patterns 4–5, Maria on both pathways. R2: Enrollment rules, KPIs for the second pathway with no code change. R3: Pathway Builder on diabetes PDTA, Enrollment Advisor rationale. R4: Nuova patologia screen	R1–R4	AC15, AC16, AC17	8
5	Bed Finder	Only if time remains after day 4. R1: `beds` and occupancy from admissions and discharges. R2: `free_beds` and ranking (diagnosis → ward, same hospital first, then distance). R3: two modes, assign (default) and propose-and-confirm, with audit and undo. R4: Posti letto screen	R1–R4	AC18, AC19	9c

Cut order if anything slips (Solution Plan §4): Bed Finder → option B → optional agents → parallel what-if → Pathway Builder (manual JSON stays) → map (comune table stays).

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9. Phase 4 — Stabilisation and demo readiness (Week 4)

Goal: the demo runs 10 times in a row, offline, on the main and the backup machine.

Day	Step	Owner	Done when
1	4.1 Feature freeze; only bug fixes merge from now on	All	Branch protection on `main`
1	4.2 Measure AC2–AC5, AC12, AC16 on the demo machine (AC5 from the cache, D18); profile and fix the slowest path. Measure peak RAM of the full app with the 6 runs and the models the demo loads	R2, R3	All within target; no swap
2	4.3 Final cache warm-up for all 3 scenarios with `qwen3:30b-instruct`; cache files hashed and copied to the backup laptop	R3	Hash list matches on both machines
2	4.4 Offline hardening: fonts (IBM Plex, OFL) and all JS in the repo; grep proves no `http` asset URLs; full run with the network adapter disabled	R4	Offline run passes
2	4.5 Record the backup video of a full run (OBS)	R4	Video on both machines
3	4.6 Rehearsals with the mentor picking the envelope; timing per step; Italian colleague reviews every on-screen term and the report. Decide whether live 4B text stays on (D14): keep it only if no rehearsal shows a slip or a wait over 10 s	R5	Under 7 min; terms approved; live-text decision recorded
4	4.7 10 consecutive runs; log every cache fall-back and stop	All	AC9: 10 of 10 without a stop
4	4.8 Final package: SPEC, code, tests, prompt log, decision register, retrospective	R5	Tagged release `demo-v1`
Show day	4.9 Runbook from Solution Plan §12: T−24h full rehearsal, T−60 min network off, mains power, cache-only on; `qwen3:30b-instruct` not loaded (the cache replaces it); `qwen3:4b-instruct` warmed with one call only if live text stays on (step 4.6), then `ollama ps` shows only that model, T−30 min projector check, T−5 min reset and start 3 scenarios	R4, R5	—

Fallbacks during the demo: the demo starts in cache-only, so no LLM call is on the critical path. If live 4B text is on and silent for 10 s → cache-only hotkey. Alarm late → jump to day 34. UI frozen → reload (state is on the server). Machine down → backup laptop, same scenario, jump to the same day. Both down → video.

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10. Zero-cost bill of materials

Need	Choice	Cost and licence note
Language and env	Python 3.12, uv	Free, open source
Data and ML	numpy, pandas, DuckDB, scikit-learn, XGBoost	Free, open source
API	FastAPI, uvicorn, WebSocket	Free, open source
Console	React, TypeScript, Vite, Node 24 via fnm	Free, open source
LLM runtime	Ollama	Free, open source
Models	Qwen 3 30B-A3B Instruct and Qwen 3 4B Instruct (Apache 2.0), D11	Free; backup Nemotron 3.5 Lightning under the NVIDIA Open Model License, which allows commercial use (not kept on disk, see D11)
Agent orchestration	Plain Python with tool calling	Replaces LangGraph: fewer dependencies
LLM call log	JSON lines in `logs/llm.jsonl` (step 2.7)	Replaces Langfuse, which needs Docker and several services
Prompt log	`docs/PROMPT_LOG.md`: useful development prompts and AI-assisted decisions (Definition of done)	Free
Map	ISTAT boundaries as GeoJSON, drawn without a basemap	No tile server, works offline
Distances	Precomputed once	No live calls to public OSM servers
Fonts	IBM Plex (OFL), bundled	No Google Fonts at runtime
Notebooks	JupyterLab	Free
Video	OBS Studio	Free
Repo	GitHub private repo	Free tier
Process mining	pm4py, notebook only, optional	Not needed for the demo
Voice input	Dropped	Typing plus the 10 benchmark questions

The only real cost is hardware. Without a GPU machine, the plan still works because of the cache, at the price of slower cache generation.

11. Cross-cutting rules

Definition of done (Solution Plan §15): code reviewed by one other role, tests pass, its AC is met, it works in the end-to-end flow, and the prompt log is updated if AI was used.

Branching: trunk-based; short-lived branches `r<role>/<topic>`; pull requests into `main` reviewed by one other role. Commit messages say what changed, not "refactor" boilerplate. No force-pushes to `main`.

Testing layers:

- Unit tests for the state engine, rules, tools, gateway and action service.
- Data tests: the pattern notebook plus asserts that run in pytest.
- Agent evaluation: 10 benchmark questions × 3 scenarios, plus the Verifier injection set, stored per prompt version.
- Performance tests: timers for AC2–AC5, AC12, AC16 on the demo machine.
- End-to-end: the full demo script, scripted through the API, every day from week 3.

Decision register (continues D1–D10 from the Solution Plan):

#	Decision	Status
D11	`qwen3:30b-instruct` for every cached LLM output (Investigator, Pathway Builder, Rule Advisor, Reporter, Verifier wording); `qwen3:4b-instruct` only for short live text that code fills from templates (action rationales, call-script wording), and only if step 4.6 keeps live text on. Both Apache 2.0, both fit in RAM together (about 21 GB of 32 GB), but the demo loads at most the 4B model. Plain `qwen3:4b` and `qwen3:30b` tags are thinking models and are not allowed. Backup: `nemotron-3.5-lightning:30b`, not kept on disk; pull it again only if both Qwen models fail. In the demo, every output from these agents is replayed from the 30B cache (D14). Evidence in §3.1	Decided 27 Oct 2026
D12	No GPU in the team: every laptop is the reference laptop (§3). One team laptop is the demo machine and generates the cache unattended (overnight runs if needed); a second identical laptop is the backup, with the same models and a hash-checked copy of the cache. Live LLM is never on the critical path	Decided 26 Oct 2026
D13	Console in React 19 + TypeScript on Vite 8, not Angular. Reasons: the prototype's screens port almost one-to-one into components; far less boilerplate for a one-week build; AI assistants generate React most reliably; the build is static files that FastAPI serves offline. Node 24 LTS via fnm (`src/ui/.node-version`), because the laptops' Node 20 is too old for Vite 8. All API paths under `/api` so one proxy rule separates API from console. Charts, Sankey and the GeoJSON map with one library chosen in week 2 (candidate: Apache ECharts, bundled locally)	Decided 26 Oct 2026
D14	Cache-first LLM: every demo LLM output is pre-generated with `qwen3:30b-instruct` and replayed. The demo starts in cache-only mode. Live 4B text for action rationales and call scripts is an optional bonus, kept only if rehearsals are clean (step 4.6). Reason: the best live investigation took 243 s against the 30 s target, and every model made at least one factual slip (§3.1)	Decided 27 Oct 2026
D15	Plain Python agents instead of LangGraph; JSON log instead of Langfuse	Proposed
D16	Offline map from ISTAT GeoJSON, no basemap; distances precomputed	Proposed
D17	Cache scope: the cache holds only LLM outputs the demo script shows (investigations, Verifier wording, rule proposal, report, Pathway Builder, and the rationales and call scripts on screen), listed in a manifest (step 2.11). Every other rationale and call script is the template filled by code, which the Solution Plan already lists as the fallback. Reason: Intervention creates actions every day for 10,000 patients in 3 scenarios; at about 11.5 s per text (§3.1), caching all of them would take days of CPU time	Proposed
D18	AC5 is measured from the cache only (under 2 s). The live target of 30 s is not reachable on CPU laptops (best measured 243 s, §3.1) and is stated in the limitations	Proposed
D19	The TL confirmed that we go ahead with CarePath AI as planned. This closes the Solution Plan §17 risk about ENG's healthcare portfolio having been sold	Decided 27 Oct 2026

12. Acceptance criteria map

AC	Criterion (short)	Phase	Owner
AC1	Agents find patterns 1–3 without hints	2	R3
AC2	Simulated day under 1 s	1 (early), 2	R2
AC3	KPI, Sankey, map under 200 ms	2	R2
AC4	What-if 6 months under 10 s	2	R2
AC5	Investigator under 2 s from cache; live 30 s not reachable on CPU (D18)	2	R3
AC6	At least 8 of 10 benchmark questions correct	2	R3
AC7	No action above its autonomy level without approval	2	R3
AC8	Fewer readmissions with agents, same seed	2	R1
AC9	10 consecutive demo runs without a stop	4	R5
AC10	Verifier catches all injected wrong numbers	3	R3
AC11	Rule Advisor rule fires alarms the next day	3	R3

AC	Criterion (short)	Phase	Owner
AC12	Worklist under 30 s; 15 of top 20 high-risk or overdue	3	R2
AC13	Correct ASST, comune group and MMG group in all 3 scenarios	1 (data), 2 (agents)	R1
AC14	Maria's trajectory identical on every run	1	R1
AC15	Enrollment finds at least 90% of pattern-5 candidates	3	R2
AC16	New pathway to compliance under 60 s	3	R2
AC17	Same tools, KPIs and console for both pathways	2	R2
AC18	Bed Finder under 3 s, correct ward, bed really free	3	R2
AC19	Confirm mode writes nothing without confirmation; assign mode audited and undoable	3	R3

13. Open questions

1. Italian colleague: no official confirmation yet of when they are available, or whether they will attend the demo. The plan still assumes their reviews of codes (step 1.3), PDTA texts (step 1.8) and on-screen terms (step 4.6).